

Lump-sum or monthly transfers for farmer support: Experimental evidence from northern Ghana*

Roxana Elena Manea^{†1}, Paul Christian^{‡2}, Felipe Alexander Dunsch^{§1},
Jonas Heirman^{¶1}, and Jennifer Waidler^{||1}

¹World Food Programme, Office of Evaluation

²World Bank, Development Impact Group

June 16, 2026

Abstract

A large literature shows that cash transfers improve welfare, but less is known about how different disbursement schedules affect household behavior and outcomes when liquidity needs and investment timing are dynamic. This paper studies how the size, frequency, and timing of transfers affect food consumption and productive decisions using experimental variation from a cash transfer targeted to farmers in northern Ghana. We implement a cluster-randomized controlled trial comparing an upfront lump-sum transfer delivered before planting to three equal installments of the same total value spaced throughout the season. Tracking households over the agricultural cycle, we show that lump-sum transfers at the start of the season induce higher agricultural investment, sales revenues, and short-run improvements in food security, while monthly transfers are associated with greater livestock accumulation.

Keywords: cash transfer design, lump-sum payment, monthly transfers, agriculture, food security, livelihoods, smallholder farmers, cluster randomized experiment

JEL Codes: I38; O12; Q12

*We use CRediT to recognize author contributions: (1) conceptualization, (2) methodology, (3) validation, (4) formal analysis, (5) investigation, (6) data curation, (7) writing original draft, (8) reviewing and editing, (9) visualization, (10) supervision, (11) project administration, and (12) funding acquisition.

[†]CRediT: 3, 4, 5, 6, 7, 8, 9, 11.

[‡]CRediT: 1, 2, 3, 4, 7, 8, 10.

[§]CRediT: 1, 2, 7, 8, 10, 11, 12.

[¶]CRediT: 1, 2, 8, 10, 12.

^{||}CRediT: 1, 2, 7, 8, 10, 11.

⁰The study is registered in the AEA RCT Registry, AEARCTR-0011394, and it was approved by Solutions IRB, Protocol#2023/05/6. The United States Aid (USAID) Agency's Bureau for Humanitarian Affairs and USAID Ghana funded this research. The WFP Country Office and WFP Field Office in Tamale have provided support in enabling the coordination of operations with the evaluation design. We thank Nidhila Adusumalli for her valuable contribution, including conducting the randomization, project coordination, and data collection supervision, as well as Jovin Lasway for research assistance. We also thank *D&D Consulting* for the quality of their surveying work.

1 Introduction

A robust body of evidence confirms that cash transfers are effective in achieving a host of development objectives. They reduce poverty, support livelihood generation, improve well-being, stimulate investments in assets and human capital, and can reduce the risk of violence (Blattman et al., 2014; Karlan et al., 2014; Bastagli et al., 2016; Tirivayi et al., 2021; Jeong and Trako, 2022; Crosta et al., 2024). As they are becoming an increasingly important part of the development toolkit, the research agenda is now focusing on decisions about when, where, and how to deliver transfers (Jeong and Trako, 2022; Crosta et al., 2024). One key design decision is timing. When both liquidity and return on investment are shifting over time, decisions about whether to deliver cash in a lump transfer or spaced out in ongoing streams may affect the use of the cash and the outcomes achieved for a given transfer size.

We report on an experimental study in a context where these considerations are important. The experiment varies the disbursement schedule of a cash transfer programme implemented by the World Food Programme and targeted at farmers in northern Ghana who face the seasonal trade-off between securing their food needs and investing in agriculture at the start of the rainy season. We show that transfer timing shapes how liquidity is allocated across agricultural inputs and other livelihood investments. Because of credit constraints and the time sensitivity of certain investments, farmers are nudged to make expenditure decisions that are best aligned with the transfer schedule that they are randomly assigned. Policy-makers may thus choose transfer schedules strategically to promote their outcomes of interest.

Specifically, the cluster randomized experiment compared two groups. One group received a lump-sum transfer just before the planting season, when liquidity was most critical for purchasing inputs (81 villages), while the other group received three monthly installments throughout the lean season (82 villages). Regardless of the disbursement method, the support amounted to the same total of USD 315 per farmer. The intervention also included specific features, such as labeling transfers for agriculture with a small food component, in line with the common practice of pairing transfer programmes with nudges and sensitization. The operational costs were equivalent across the two distribution schemes.

We conducted three survey rounds timed to transfer delivery: after the lump-sum transfer (while the comparison group had received only one installment), after completion of the monthly transfers, and at the end of the harvest season. In contrast, most studies rely on baseline and endline snapshots alone, which can miss short-lived but meaningful dynamics (Crosta et al., 2024; Aggarwal et al., 2024).

This study shows that the randomized distribution schedule has shaped how farmers allocated resources. Beneficiaries in the lump-sum group used 20% more household labor (4.4 additional days at harvest time) and spent 7.4% more on chemical fertilizer (GHS 120). They also cultivated 5.5% more land (0.33 acres), though this last estimate is less precise. These input differences translated into higher crop revenues: lump-sum households earned significantly more, i.e., 16% larger crop sales (86 Kg and GHS 283). In contrast, monthly transfers were associated with greater livestock accumulation: the Tropical Livestock Units (TLU) were 0.18 higher for the households receiving monthly transfers, leading to a mean TLU of 1.31, especially owing to large cattle and sheep.

Moreover, households receiving lump-sum transfers experienced an early advantage in food security. As of round 1 of data collection, their Food Consumption Score (FCS) was 2.1 points higher than the monthly group’s mean of 59.8, and their reduced Coping Strategies Index (rCSI) was 2 points lower relative to the monthly group’s mean of 19.9, indicating better dietary diversity and less frequent negative coping behaviors. By rounds 2 and 3, however, the lump-sum group’s initial food security advantage had reduced: FCS differences flipped slightly (-1.05 and -0.59), and rCSI gaps narrowed to near zero, reflecting convergence, as monthly transfer recipients completed their installments. Importantly, however, the lump-sum group’s short-term boost in food security happened without significant deterioration later.

Furthermore, the effects of varying the payment schedule were not uniform. The literature has shown that farm size mediates the impact of transfers (Dercon and Christiaensen, 2011; Fafchamps et al., 2014). Similarly, we consider farm size to be a proxy for wealth and hypothesize that wealthier (larger) farms are better able to buy fertilizer. We find suggestive evidence that households with smaller farm operations, likely facing tighter liquidity constraints, benefited disproportionately from lump support, both in terms of input use and early-season food security, while larger farms showed little sensitivity across modalities. For instance, fertilizer spending for lump-sum recipients was significantly higher at the 5th and 25th percentiles (GHS 176 and GHS 165, respectively), relative to the average of GHS 120.

This experiment thus contributes to a growing literature on design decisions that influence the impacts of cash transfers. For instance, studies have compared conditional to unconditional support (Baird et al., 2010), cash to in-kind transfers (Fafchamps et al., 2014; Hidrobo et al., 2014; Hoddinott et al., 2018; Schwab, 2019), explored the role of labeling (Benhassine et al., 2015), digital payments (Aker et al., 2016), and assessed targeting strategies (Alatas et al., 2012; Karlan and Thuysbaert, 2019; Premand and Schnitzer, 2021). Other work also highlights learning from gender-sensitive targeting (Haushofer et al., 2019; Christian et al., 2023; Balantrapu et al., 2025) as well as the use of machine learning to improve targeting efficiency (McBride and Nichols, 2018; Aiken et al., 2022;

Haushofer et al., 2025; Baliki et al., 2026). Banerjee et al. (2015) further show that *cash+* programs that combine transfers with training and assets produced lasting gains, though at a higher cost. Similarly, Karlan et al. (2014) argue that capital and risk constraints co-exist, and transfers bundled with insurance bring greater effects. In contrast, McIntosh and Zeitlin (2022) state that cash alone can spark sustained growth when targeted appropriately, and training programs add limited value relative to cost-equivalent transfers.

Despite the aforementioned growing literature that compares alternative transfer designs, evidence on the implications of varying the size, frequency, and timing of disbursements remains limited. This study contributes to narrowing this research gap and is closest in scope to Haushofer and Shapiro (2016).

Several theoretical mechanisms help explain why outcomes may differ across disbursement schedules. Poverty-trap models emphasize the presence of thresholds in investment, whereby sufficiently large transfers enable households to undertake lumpy investments that generate sustained income gains (Banerjee et al., 2015; Bedoya et al., 2019; Balboni et al., 2022; Aggarwal et al., 2024). Additionally, when returns to investment vary seasonally, as in rain-fed agriculture, the timing of liquidity provision, in addition to the liquidity itself, also becomes critical. Consequently, one-off well-timed transfers are well positioned to enable time-sensitive lumpy expenditure decisions.

In contrast, alternative schedules such as monthly installments are generally calibrated to smooth consumption and meet basic needs on a monthly basis (Berhane et al., 2014; Haushofer and Shapiro, 2016).

Emerging evidence does suggest that larger, less frequent payments may help households to better overcome liquidity constraints and invest in productive assets (Bastagli et al., 2016), while not excluding concurrent improvements in food security (Banerjee et al., 2020). At the same time, however, there is also evidence that points to diminishing marginal returns (Kondylis and Loeser, 2024), and meta-analyses suggest no consistent advantage of very large one-time transfers over moderate amounts (Crosta et al., 2024).

No single disbursement structure is expected to dominate universally. Welfare effects depend on how transfer timing interacts with liquidity constraints, investment lumpiness, and the temporal profile of returns. We argue that in the context of smallholders engaged in rain-fed agriculture, transfer size, frequency, and timing emerge as core design parameters rather than administrative details. Lump-sum transfers are better suited to environments with seasonal, lumpy investment opportunities, while monthly transfers align more closely with recurrent consumption needs and gradual asset accumulation. Abstracting from these distinctions risks attenuating the welfare and productivity impacts of otherwise cost-equivalent transfers.

2 Background

Ghana’s socioeconomic development has been relatively slower in the north as compared to the more dynamic south, where poverty reduction has been more pronounced.¹ Although the agricultural sector is key to the region’s economic welfare², rising agricultural input prices³ and low soil fertility⁴ have discouraged smallholder farmers from engaging in agricultural activities. The violence in Burkina Faso, which led to refugees seeking shelter in northern Ghana as well as in other neighboring countries, has also put additional strain on the north’s limited resources. High food inflation⁵ and a generally slow economic recovery have further motivated the need to support small-scale farmers in these relatively worse-off and more vulnerable areas of the country.

Against this background, the World Food Programme (WFP), with support from the former U.S. Agency for International Development (USAID), implemented the Farmer Support Activity (FSA) in an effort to ease the constraints faced by smallholder farmers in northern Ghana. The FSA was implemented during the 2023 agricultural season and consisted of transferring USD 315 (GHS 3,465) in mobile money as a farming grant.

The FSA support represented roughly 13 percent of Ghana’s GDP per capita in 2023.⁶ The size of the transfer was set to cover the estimated expenditure with agricultural inputs for 1 hectare of land (~ 86 percent of the FSA amount, including an accompanying government subsidy), as well as a portion of the households’ food expenditure (~ 14 percent). Thus, the smallest of farms, representing 27 percent of the sample, would have seen most of their agriculture-related expenditures covered entirely.

¹<https://pip.worldbank.org/country-profiles/GHA>

²Despite being endowed with relatively less fertile land, the northern regions of Ghana (Northern, North East, Upper East, and Upper West) produce 95 percent of the country’s millet output. Similarly, the same regions account for 86 percent of the sorghum and soy productions, 79 percent of the cowpea output, 70 percent of groundnuts, 32 percent of yams, and 25 percent of the country’s maize production. These estimations are per the most recently released 2022 statistics (Ghana Statistical Service, 2022).

³According to the Ministry of Food and Agriculture and World Food Programme (2023), the price of the compound fertilizer NPK 15 15 15 increased by 12 percent in July 2023 relative to July 2022 and by 127 percent relative to its five-year average price. Similarly, the percentage increases for sulfate of ammonia were 17 and 131, respectively. For urea, the price increased by 22 percent relative to 2022 and 142 percent relative to the five-year average. More details about input prices can be found in the cited situation report, also available at <https://reliefweb.int/node/3998612>.

⁴According to the Ministry of Food and Agriculture (2019), soil fertility in northern Ghana relative to other regions is shown to be poorer in tables 2.7, 2.8, and appendix 12 in Ghana’s yearly facts and figures about the agricultural sector. Bationo et al. (2018) further argue that Ghana’s soils are generally poor relative to other countries in the region and face high nutrient depletion.

⁵According to the Ghana Statistical Service (2024), the yearly food inflation in June 2023 was 54 percent. For February 2024, it was at 27 percent. Relative to 2021, the Consumer Price Index was 184 percent as of June 2023 and 208 percent as of February 2024. Moreover, inflation in the northern regions of Ghana was generally higher than the national average.

⁶<https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=GH>

WFP’s usual approach has been to make regular transfers of equal size to help beneficiaries smooth their consumption during difficult times. For the FSA, however, an alternative approach was considered: making a lump-sum transfer to disburse most of the cash upfront, when it was most needed for agricultural investment, also considering the north’s reliance on one main agricultural season.

3 Experimental Design

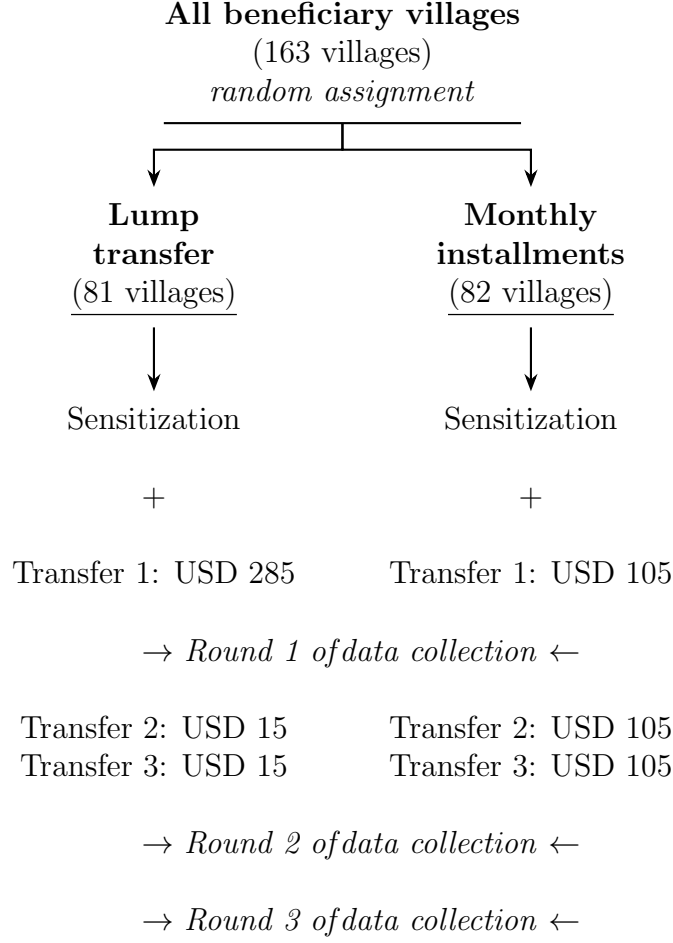
The FSA was present in 163 villages across 17 districts in 4 of Ghana’s regions: Northern, North East, Upper East, and Upper West. All participating villages were considered for the randomized assignment. Consequently, half of them were selected to receive lump-sum support, and the remaining half benefited from monthly transfers (Figure 1). The experiment did not include a pure control group. As argued previously, the added value of this study lies in refining *how* transfers may be delivered in similar contexts. The design of choice thus aligns with the growing literature that is shifting from evaluating whether something works to how it could work better, and whether there are subgroups of targeted populations that benefit disproportionately. The randomized assignment was clustered at the village level and stratified at the district level. Thus, all beneficiary households within any one village received their transfers according to the same schedule.

The A/B type of experimental testing limits the study’s capacity to assess the impact of the FSA on smallholder farmers, but it enables the estimation of the comparative advantages of each disbursement schedule relative to the other.

The FSA support was largely designed to cater to the agriculture-related needs of smallholder farmers, e.g., fertilizer, seeds, and mechanization support. Additionally, a small food component was embedded and labeled as such to prevent farmers from making trade-offs between their food and livelihood needs. Briefly, farmers were told the FSA grant was for purchasing agricultural inputs, and that it also included a small component of approximately 15 USD per transfer for their food-related expenditure. The support was not conditioned as such, but the messaging was meant to nudge farmers to invest in agriculture and spend a small portion of their grants on food.⁷

⁷Only 3 percent of household instances indicated problems with the receipt of their mobile money, and the issue was balanced across the two comparison groups. Importantly, WFP’s monitoring data has also indicated that there were no recorded discrepancies between the observed disbursement modality and the assigned schedule at the village level. However, given that some beneficiaries had limited prior experience with mobile money, it is plausible to assume that not all farmers have immediately become aware of the receipt of their transfers and accessed them in a timely manner. As a result, some may have behaved as though they were subject to a different disbursement schedule than the one assigned to them. The primary risk posed by this type of partial compliance consisted of attenuating the estimated effects: either by masking the true size of differential impacts or making them harder to detect.

Figure 1: Experimental design



The first of the three FSA transfers was planned to happen in June, when the application of fertilizer was timely. Thereafter, the remaining transfers were scheduled to occur throughout the lean season. Data collection was then scheduled such that information would be gathered: (1) after the first transfer took place (July 2023), (2) after all transfers took place (October 2023), and (3) at the end of the agricultural season (February 2024) (Figure 1).⁸ Surveys were collected on tablets during face-to-face discussions.

Appendix Table A1 uses registration data to show that the randomization process was successful and there are no notable statistically significant differences between the two comparison groups. Particularly, we examined various demographic characteristics, agricultural activities, and financial aspects. Importantly, there is also no evidence of a significant differential non-response rate among the two groups, as the difference is small, standing at less than 2 percentage points.

⁸Table A2 provides a summary of the outcomes of interest and the round when they were collected, as well as the sample size for each round.

4 Estimation

We assess the lump-monthly differential impacts for the pooled dataset as indicated in Equation 1.

$$y_{fvr} = \alpha + \beta t_v + \sum_{r=2}^3 \delta_r r_r + \sum_{s=2}^{17} \gamma_s s_s + \sum_{r=2}^3 \sum_{s=2}^{17} \eta_{sr} s_s r_r + \epsilon_{fvr} \quad (1)$$

Where y_{fvr} is the outcome variable for farmer f from village v in round r . α is a constant. The variable denoting the treatment arm is t_v . It varies at the village level, taking the value of 1 if the village benefits from lump support and 0 otherwise. The coefficient of interest is β , i.e., the estimated impact of receiving the lump-sum transfer as opposed to three regular monthly payments. Then, r_r as well as s_s are dummy variables for rounds 2 and 3, and for each of the 16 districts where treatment at the village level was stratified. Round 1 and district 1 are omitted as base categories. The model also interacts the round with the district of data collection to further account for unobservables that are district-round specific.

To assess whether impacts may differ in size across rounds, we also estimate Equation 1 one round at a time as indicated in Equation 2. We hypothesize that the timing and size of transfers matter, and thus, impacts are unlikely to be the same if measured at various intervals of time relative to the timing of transfers or of the harvest season, especially as far as food security outcomes are concerned.

$$y_{fv}^{(r)} = \alpha + \beta t_v + \sum_{s=2}^{17} \gamma_s s_s + \varepsilon_{fv}^{(r)}, \text{ for } r = 1, 2, 3 \quad (2)$$

Finally, we explore the heterogeneity of impacts with respect to farm size using the model in Equation 3 for the pooled sample and Equation 4 for each round separately.

$$y_{fvr} = \alpha + \beta_1 t_v + \beta_2 (t_v \times h_f) + \theta h_f + \sum_{r=2}^3 \delta_r r_r + \sum_{s=2}^{17} \gamma_s s_s + \sum_{r=2}^3 \kappa_r (r_r \times h_f) + \sum_{s=2}^{17} \omega_s (s_s \times h_f) + \sum_{r=2}^3 \sum_{s=2}^{17} \eta_{sr} s_s r_r + \nu_{fvr} \quad (3)$$

$$y_{fv}^{(r)} = \alpha + \beta_1 t_v + \beta_2 (t_v \times h_f) + \theta h_f + \sum_{s=2}^{17} \gamma_s s_s + \sum_{s=2}^{17} \omega_s (s_s \times h_f) + \nu_{fv}^{(r)}, \text{ for } r = 1, 2, 3 \quad (4)$$

Where h_f is a continuous variable denoting farm size. We hypothesize that the smallest of farms are the ones most constrained by their lack of access to liquidity. Thus, it is for them that the timing and size of transfers are likely to matter the most.

5 Results

Results are grouped into 4 categories: food security and well-being (Section 5.1), crop activities, including inputs and outputs (Section 5.2), other livelihood strategies such as livestock rearing, business, and wage labor engagements (Section 5.3), and assets and financial tools (Section 5.4). Finally, we assess farm size heterogeneity across the main outcomes of interest in Section 5.5. Results are reported by data collection round to reflect impacts at key moments in the distribution cycle, and then we also show the pooled findings in the last panel of each table.

5.1 Food security and well-being

Table 1 presents the differential impacts of the two disbursement schedules as they relate to food security and mental well-being outcomes. The selected indicators are described in detail in Appendix A.2. Briefly, a higher Food Consumption Score (FCS) suggests a greater likelihood that a household’s food intake meets adequacy standards. In contrast, the reduced Coping Strategies Index (rCSI) and the Food Insecurity Experience Scale (FIES) capture food *insecurity*: rCSI serves as a proxy for the intensity (i.e., frequency) of coping behaviors adopted in response to food shortages, while FIES reflects the presence of such experiences (summed binary assessments). Then, the Patient Health Questionnaire for Depression and Anxiety (PHQ-4) is an ultra-brief screening tool for psychological distress, with scores categorized as normal (0–2), mild (3–5), moderate (6–8), or severe (9–12).

Panels A–C indicate that households receiving lump-sum transfers experienced better food security immediately after the first disbursement. For example, the estimated FCS gap ($\beta = 2.10$, $p < 0.10$) suggests that these households consumed, on average, two additional days of vegetables or one extra day of staples as compared to those receiving monthly payments. A similar pattern emerges for coping strategies: in round 1, lump-sum households were significantly less likely to resort to negative coping mechanisms, as reflected by a lower rCSI ($\beta = -1.997$, $p < 0.01$), equivalent to roughly two fewer days of relying on less preferred foods in the past week, for instance. These effects fade in subsequent rounds. By rounds 2 and 3, monthly-payment households slightly outperform lump-sum recipients, though differences are small and statistically insignificant at conventional levels.

Table 1: Food security and mental well-being

	(1) FCS (7 days)	(2) rCSI (7 days)	(3) FIES (30 days)	(4) PHQ4 (2 weeks)
Panel A: Round 1				
Lump-sum payment	2.100*	-1.997***	-0.233	-0.144
District dummies	Yes	Yes	Yes	Yes
Mean monthly group	59.81	19.89	5.61	4.45
Observations	3237	3237	3237	3237
Panel B: Round 2				
Lump-sum payment	-1.049	0.225	-0.102	0.038
District dummies	Yes	Yes	Yes	Yes
Mean monthly group	55.91	13.52	4.80	10.26
Observations	2993	2993	2993	2993
Panel C: Round 3				
Lump-sum payment	-0.591	0.048	-0.023	0.108
District dummies	Yes	Yes	Yes	Yes
Mean monthly group	49.19	8.89	3.24	3.81
Observations	3054	3054	3054	3054
Panel D: Pooled				
Lump-sum payment	0.199	-0.607	-0.122	-0.003
Round dummies	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes
Mean monthly group	55.07	14.23	4.57	6.11
Observations	9284	9284	9284	9284

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). Measures of food security (Food Consumption Score), food *insecurity* (reduced Coping Strategies Index, Food Insecurity Experience Scale), stress and anxiety (Patient Health Questionnaire 4). The recall period for each measure is mentioned in the header between parentheses.

Panel D, which pools all rounds, shows that these early advantages become diluted as the group receiving monthly transfers catches up: the estimated coefficients for lump-sum payments are close to zero and statistically insignificant across all indicators. This, however, occurs despite the lump-sum group receiving only a small top-up (USD 15) while the monthly group continued to receive USD 105. As the monthly group does not outperform lump-sum recipients significantly, this pattern may indicate effective consumption smoothing also among lump-sum households, potentially reinforced by programme messaging and sensitization efforts.

Finally, the disbursement modality appears unrelated to mental health outcomes, as measured by PHQ-4 across all rounds. We re-emphasize that this is a comparison across modalities and not an assessment of programme impacts, which might have led to different insights, especially in light of Haushofer and Shapiro (2016).

5.2 Crop cultivation activities

The information on crop cultivation was collected once in round 3, at the tail end of the harvest season. Table 2 shows that farmers who received lump-sum support mobilized about 20% more household labor and spent approximately 7.4% more on chemical fertilizers as compared to households receiving monthly transfers. They also cultivated slightly more land: an increase of around 5.5%. Although the effect on the area cultivated is not statistically significant in this specification, it remains economically meaningful.⁹

Table 2: Inputs for the main agricultural season

	(1) Area worked (Acres)	(2) Household labor (Days Feb.)	(3) Chemical fertilizer (Yes/No)	(4) Chemical fertilizer (GHS)	(5) Organic fertilizer (Yes/No)	(6) Organic fertilizer (GHS)
Panel A: Inputs						
Lump-sum payment	0.327 (0.244)	4.391** (2.217)	0.015 (0.011)	120.331** (59.759)	0.005 (0.013)	0.664 (8.551)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	5.90	22.07	0.90	1628.64	0.08	42.47
Observations	3054	3054	3054	3054	3054	3054
	Seeds (Yes/No)	Seeds (GHS)	Pesticides (Yes/No)	Pesticides (GHS)	Hired labor (Yes/No)	Hired labor (GHS)
Panel B: Inputs (continued)						
Lump-sum payment	-0.004 (0.021)	14.044 (23.936)	0.011 (0.011)	-1.673 (20.562)	-0.010 (0.022)	-10.407 (18.379)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.71	361.16	0.94	396.57	0.66	250.73
Observations	3054	3054	3054	3054	3054	3054

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). All continuous variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. Data on the inputs for the 2023 rainy season were collected once during round 3. Yes/No answers are binary coded 1/0.

The more intensive use of agricultural inputs, land, and labor also translates into a larger harvest, i.e., 6 percent more than the average quantity harvested by the monthly transfers group, although this difference is not statistically significant (Table 3). However, households receiving lump-sum transfers report significantly higher revenue from crop sales, with revenues ~ 16 percent greater than those of the monthly transfers group. These findings show that lump-sum transfers enabled the channeling of significantly more resources toward crop cultivation as compared to the alternative payment schedule.

⁹When estimating a log-normal function instead of winsorizing the dependent variable (see Appendix Table A4), the coefficient becomes statistically significant, suggesting that with a larger sample size, the positive effect would likely be more robust.

Table 3: Outputs from the main agricultural season

	(1) Harvest (GHS)	(2) Harvest (Kg)	(3) Sales (Yes/No)	(4) Sales (Kg)	(5) Sales (GHS)
Lump-sum payment	399.787 (393.385)	117.941 (107.519)	0.029 (0.024)	85.524** (42.460)	282.995* (155.959)
District dummies	Yes	Yes	Yes	Yes	Yes
Mean monthly group	6877.36	1837.15	0.71	469.11	1755.92
Observations	3054	3054	3054	3054	3054

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). All continuous variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. Data on the outputs for the 2023 rainy season were collected once during round 3. All crops that are reported at least once are considered. Grains and pulses are the main types of crops that farmers report (i.e., maize, soya, rice, millet, and beans are reported by 93, 59, 30, 20, and 19 percent of farmers). The most frequently reported non-cereal and non-pulse crops are groundnuts and yams (37 and 10 percent, respectively).

5.3 Other livelihood strategies

While agriculture is the main source of livelihood for households in the sample, we also look at other sources of income, including livestock rearing, business activities, and wage employment to ascertain whether there are further differential impacts among the two disbursement schemes.

Table 4 presents the regression estimates for the livestock rearing outcomes. The disbursement schedule had no significant effect on the likelihood of rearing livestock, nor did it influence profits from livestock sales or the value of the consumed livestock. However, when livestock ownership is measured using the Tropical Livestock Unit (TLU) aggregator, we observe positive and statistically significant effects in favor of the group receiving monthly transfers. This effect is particularly owed to increases in the number of cows and sheep that households own (Appendix Table A7).¹⁰

Beyond livestock, Table 5 indicates that household business activities remained largely unaffected by the disbursement modality. Whether transfers were provided as lump support or in monthly installments, households reported comparable rates of business ownership and engagement. Business engagement was measured by the number of businesses operated as well as the total labor days contributed by household members in the past month.

¹⁰The TLU is computed based on each household's stock of cows, pigs, sheep, goats, and poultry using the FAO guidelines in terms of the applicable weights. <https://www.fao.org/4/t0828e/T0828E07.htm#ch5>. The equation is: $TLU = 1 \times cows + 0.2 \times pigs + 0.1 \times sheep + 0.08 \times goats + 0.01 \times poultry$, whereby poultry includes chickens, ducks, fowls, pigeons, etc. Data was not collected for donkeys or horses explicitly. There were a few instances of such livestock, but due to their reporting as "other" livestock, which can include further types of livestock, they cannot be disentangled and thus are not included in the TLU computation.

Table 4: Livestock rearing

	(1) Reared livestock (Yes/No) (12 months)	(2) TLU (Current)	(3) Profits livestock (GHS) (6 months)	(4) Consumed livestock (GHS) (6 months)
Panel A: Round 2				
Lump-sum payment	0.012 (0.022)	-0.169* (0.098)	15.196 (16.864)	3.077 (14.285)
District dummies	Yes	Yes	Yes	Yes
Mean monthly group	0.73	1.29	142.30	134.83
Observations	2993	2993	2993	2993
Panel B: Round 3				
Lump-sum payment	-0.015 (0.016)	-0.183* (0.099)	-5.864 (14.085)	5.878 (12.959)
District dummies	Yes	Yes	Yes	Yes
Mean monthly group	0.81	1.33	146.38	126.68
Observations	3054	3054	3054	3054
Panel C: Pooled				
Lump-sum payment	-0.002 (0.016)	-0.176** (0.086)	4.561 (11.906)	4.492 (10.575)
Round dummies	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes
Mean monthly group	0.77	1.31	144.36	130.72
Observations	6047	6047	6047	6047

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). All continuous variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. Livestock data was not collected in round 1. If a household does not report livestock, then the outcomes related to livestock are coded zero. The winsorized TLU uses the winsorized stock of animals and applies the appropriate weights before summing up to compute TLU.

Table 5: Business activities and wage employment among household members

	(1) Business engagement (Yes/No) (12 months)	(2) Business engagement (Count) (12 months)	(3) Business household labor (Days) (30 days)	(4) Business profit (GHS) (30 days)	(5) Wage members employed (Yes/No) (12 months)	(6) Wage members employed (Yes/No) (last 30 days)	(7) Wage members employed (Count) (12 months)	(8) Wage members employed (Count) (last 30 days)	(9) Wage days worked (Count) (12 months)	(10) Wage yearly earnings (GHS) (12 months)
Panel A: Round 2										
Lump-sum payment	-0.006 (0.025)	-0.016 (0.034)	-0.408 (0.862)	-0.706 (10.832)	0.009 (0.022)	0.013 (0.019)	0.045 (0.045)	0.022 (0.038)	-2.790 (5.668)	-117.497 (282.886)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.31	0.37	7.66	80.29	0.31	0.23	0.53	0.37	55.21	2205.80
Observations	2993	2993	2993	2906	2993	2993	2993	2993	2993	2993
Panel B: Round 3										
Lump-sum payment	-0.026 (0.023)	-0.027 (0.030)	-0.909 (0.943)	-9.353 (12.659)	0.009 (0.021)	-0.005 (0.018)	0.026 (0.040)	-0.003 (0.027)	0.982 (5.585)	169.624 (284.035)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.42	0.48	12.50	119.82	0.28	0.19	0.42	0.25	41.96	1733.68
Observations	3054	3054	3054	2990	3054	3054	3054	3054	3054	3054
Panel C: Pooled										
Lump-sum payment	-0.016 (0.022)	-0.022 (0.029)	-0.661 (0.821)	-5.090 (10.075)	0.009 (0.017)	0.004 (0.014)	0.035 (0.033)	0.009 (0.024)	-0.885 (4.327)	27.496 (228.897)
Round dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.36	0.42	10.10	100.28	0.30	0.21	0.47	0.31	48.52	1967.63
Observations	6047	6047	6047	5896	6047	6047	6047	6047	6047	6047

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). Continuous variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. There is a considerable amount of true missing for the profit variable, as households could not estimate profits and answered “don’t know”. The number of observations is thus lower in column (4). The rate of “don’t know” is similar across the two groups: 74 vs. 77 observations in the monthly and lump group. The outcomes related to the number of days worked and yearly wage are computed at the household level by summing up the individual outcomes of household members. Members aged 12 or above were asked about their employment status. A day is counted as a workday regardless of the number of hours worked during that day. Wages are estimated for the entire year based on the reported number of months, weeks, and days a person worked and the most recent wage rate they declared. If a second wage source is reported, then both sources serve to compute the yearly workload (days) and wage, respectively.

Furthermore, Table 5 shows that differences in business profits between the two groups are neither statistically significant nor economically meaningful. The disbursement scheme variation also led to no changes in the households' participation in wage employment at either the extensive or intensive margins.

5.4 Assets and access to financial tools

Lump-sum transfers also did not lead to meaningful differences in household or farm asset ownership relative to the receipt of monthly transfers (Table 6). Point estimates for both outcomes are small and statistically insignificant.

Table 6: Assets, savings, and credit

	(1) Farm assets (Number)	(2) Household assets (Number)	(3) Balance savings (GHS)	(4) Balance credit (GHS)
Panel A: Round 2				
Lump-sum payment	0.129 (0.255)	-0.464 (0.429)	-24.069* (14.363)	-34.699 (36.188)
District dummies	Yes	Yes	Yes	Yes
Mean monthly group	8.40	11.67	150.07	283.75
Observations	2993	2993	2993	2993
Panel B: Round 3				
Lump-sum payment	-0.047 (0.240)	-0.264 (0.383)	-15.923 (17.747)	-38.605 (35.016)
District dummies	Yes	Yes	Yes	Yes
Mean monthly group	8.87	13.42	221.73	280.89
Observations	3054	3054	3054	3054
Panel C: Pooled				
Lump-sum payment	0.040 (0.236)	-0.363 (0.394)	-19.955 (13.089)	-36.671 (28.254)
Round dummies	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes
Mean monthly group	8.64	12.55	186.22	282.31
Observations	6047	6047	6047	6047

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at village level (163 clusters). Outcomes are measured at the household level. Continuous variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. Credit includes formal and informal sources.

In terms of financial outcomes, lump-sum households showed a slightly lower savings balance on average but not significant at conventional levels (Panel C, Table 6). In round 2, however, the balance was significantly lower in the lump-sum group (Panel A), but the gap reduced in round 3 in absolute terms as well as relative to the average savings of the monthly group households (Panel C). Self-reported data on transfer allocations suggests

that this may be due to households in the lump-sum group saving less and channeling more of the transfer toward fertilizer purchases, for instance. Credit balances were also consistently lower for lump-sum households, although these differences are small and not significant. Taken together, the pattern of lower savings and borrowing suggests that lump-sum recipients may have relied more heavily on the transfer itself to fund purchases given its size and timing; they had liquidity upfront, so the need to save or borrow later was reduced.

5.5 Farm scale heterogeneity

We examine in this section whether the effects of changing the transfer schedule vary according to farm scale, as declared before the randomized assignment. We use farm size as a proxy for wealth and likely less stringent liquidity constraints. We look at food security and agriculture, as these were the main outcomes of interest. Other outcomes are available in the Appendix.

In the first follow-up round, lump-sum recipients reported significantly higher food consumption scores and lower reliance on coping strategies, but these differences faded in subsequent rounds (Table 1). Heterogeneity by farm size (Table 7) shows that the gains in round 1 concentrated among smaller and medium-sized farms: effects were strongest at the 5th percentile (1.5 acres), remained significant at the 25th and 50th percentiles, and became null at the top of the distribution (95th percentile). Overall, however, the evidence points to partial heterogeneity: lump-sum transfers appear to benefit smaller farms more, but the pattern is neither systematic nor persistent and dominant, i.e., the interaction term is not significant.

Regarding agricultural outcomes (Table 8), spending on chemical fertilizers was consistently and significantly higher among smaller and medium-sized farms, with no detectable differences for larger farms. Labor use showed some increases among medium-to-larger farms, but these effects were small and only marginally significant. Finally, there were no other systematic effects regarding the area cultivated or the purchased seeds.

The higher fertilizer use among lump-sum smaller farms coincided with increases in harvest quantities and values, though these estimates are imprecise and not statistically significant. Their magnitudes, however, are noteworthy: almost 140 extra kilograms for the smallest farms receiving lump support. Sales outcomes show a clearer pattern: while the quantities sold did not vary significantly by farm size, sales revenues were significantly higher for lump-sum households at the lower and median percentiles, suggesting perhaps their choice to sell more valuable crops.

Table 7: Farm scale heterogeneity and food security

	Round 1			Round 2			Round 3			Pooled		
	(1) FCS (7days)	(2) rCSI (7 days)	(3) FIES (30 days)	(4) FCS (7days)	(5) rCSI (7 days)	(6) FIES (30 days)	(7) FCS (7days)	(8) rCSI (7 days)	(9) FIES (30 days)	(10) FCS (7days)	(11) rCSI (7 days)	(12) FIES (30 days)
Lump-sum payment	2.910* (1.540)	-2.214** (0.894)	-0.311* (0.184)	-1.408 (1.295)	-0.152 (0.742)	-0.124 (0.188)	0.947 (1.320)	-0.226 (0.672)	-0.131 (0.176)	0.892 (1.090)	-0.906 (0.554)	-0.192 (0.125)
Farm scale registered (acres)	0.038 (0.316)	0.120 (0.191)	0.046 (0.049)	-0.377 (0.303)	0.198 (0.182)	0.057 (0.050)	0.099 (0.227)	-0.288 (0.207)	-0.074 (0.054)	-0.061 (0.221)	0.015 (0.133)	0.027 (0.021)
Lump-sum × Farm scale	-0.176 (0.220)	0.070 (0.144)	0.020 (0.025)	0.050 (0.191)	0.098 (0.120)	0.009 (0.028)	-0.374** (0.155)	0.071 (0.092)	0.027 (0.025)	-0.175 (0.132)	0.082 (0.086)	0.019 (0.017)
Round dummies	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District x Round dummies	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes
District x Farm scale	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Round x Farm scale	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Observations	3237	3237	3237	2993	2993	2993	3054	3054	3054	9284	9284	9284
Marginal means at various percentiles farm scale												
Lump-Monthly at 5th perc.	2.646**	-2.109***	-0.281*	-1.333	-0.006	-0.11	.385	-0.12	-0.09	.629	-.782*	-.164
Lump-Monthly at 25th perc.	2.558**	-2.074***	-0.271*	-1.308	.043	-0.106	.198	-0.084	-0.076	.541	-.741	-.154
Lump-Monthly at 50th perc.	2.205**	-1.933***	-0.231	-1.208	.238	-0.088	-.55	.057	-0.021	.191	-.577	-.116
Lump-Monthly at 75th perc.	2.029*	-1.863***	-0.211	-1.158	.336	-.08	-.924	.128	.006	.016	-.494	-.097
Lump-Monthly at 95th perc.	1.148	-1.512	-.111	-.909	.824	-.036	-2.795**	.482	.143	-.861	-.083	-.002

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). The recall period for each measure is between parentheses. In terms of farm size distribution: 1.5 acres (5th percentile), 2 acres (25th percentile), 4 acres (50th percentile), 5 acres (75th percentile), and 10 acres (95th percentile).

Table 8: Farm scale heterogeneity and crop cultivation

	(1) Area cultivated (Acres)	(2) Pooled labor (Days Feb.)	(3) Chemical fertilizer (GHS)	(4) Seeds (GHS)
Lump-sum payment	0.546 (0.354)	3.114 (2.877)	211.948** (83.892)	-8.803 (32.794)
Farm scale registered (acres)	0.171** (0.072)	0.857 (1.166)	9.937 (22.495)	9.445 (6.558)
Lump-sum $\times$ Farm scale	-0.067 (0.065)	0.260 (0.476)	-23.635* (14.079)	5.182 (6.027)
Observations	3054	3054	3054	3054
Marginal means at various percentiles farm scale				
Lump-Monthly at 5th perc.	.445	3.504	176.495**	-1.029
Lump-Monthly at 25th perc.	.411	3.634	164.678**	1.562
Lump-Monthly at 50th perc.	.277	4.154*	117.407**	11.927
Lump-Monthly at 75th perc.	.209	4.415*	93.772	17.109
Lump-Monthly at 95th perc.	-.127	5.715	-24.405	43.021
	Harvest all crops (Kg)	Harvest all crops (GHS)	Sales all crops (Kg)	Sales all crops (GHS)
Lump-sum payment	169.527 (117.298)	588.970 (448.132)	75.571 (46.600)	363.267* (195.883)
Farm scale registered (acres)	28.675 (28.071)	123.297 (110.640)	2.948 (9.544)	9.375 (29.863)
Lump-sum $\times$ Farm scale	-19.885 (23.901)	-72.422 (85.715)	-0.732 (9.213)	-27.320 (36.264)
District dummies	Yes	Yes	Yes	Yes
District $\times$ Farm scale	Yes	Yes	Yes	Yes
Observations	3054	3054	3054	3054
Marginal means at various percentiles farm scale				
Lump-Monthly at 5th perc.	139.7	480.337	74.473*	322.287*
Lump-Monthly at 25th perc.	129.757	444.126	74.107*	308.627*
Lump-Monthly at 50th perc.	89.988	299.282	72.643*	253.988*
Lump-Monthly at 75th perc.	70.103	226.86	71.911*	226.669
Lump-Monthly at 95th perc.	-29.321	-135.249	68.25	90.071

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). All continuous variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. Data on the 2023 agricultural rainy season were collected once during round 3. In terms of farm size distribution: 1.5 acres (5th percentile), 2 acres (25th percentile), 4 acres (50th percentile), 5 acres (75th percentile), and 10 acres (95th percentile).

Taken together, the results show that lump-sum transfers primarily benefited the smaller landholders. These households appear to channel their lump-sum support into short-term consumption smoothing as well as input purchases, which translated into higher food security and increases in crop production and sales. Larger farms, by contrast, were not sensitive to the disbursement scheme. This can be explained by smallholders being more cash-constrained and therefore better positioned to leverage lump-sum payments for immediate consumption and small-scale productive investments.

6 Conclusion

This study compared two cash transfer schedules benefiting farmers in northern Ghana: an upfront lump sum *versus* three equal monthly installments of the same total value. Importantly, the two schedules did not incur different operational costs.

Households receiving the lump support experienced higher short-term food consumption and lower reliance on negative coping strategies. These differences were concentrated among the smaller farms, but largely faded during later rounds of data collection, once both groups received their full support. Importantly, the lump-sum group never became significantly worse off, suggesting thus a net welfare gain in terms of food security relative to monthly transfers.

In terms of agricultural outcomes, lump-sum transfers facilitated greater input use, particularly in terms of chemical fertilizer, as compared to monthly installments. Moreover, this pattern, too, was led by the smaller farms, likely facing more significant liquidity constraints, which were best relaxed by accessing the lump transfer at an opportune time. The higher input use among the beneficiaries of lump-sum support translated into higher crop sales: they sold significantly larger quantities and achieved higher sales revenues. Even though harvest estimates were not precise, they were large enough in terms of economic magnitude.

While lump-sum transfers yielded clearer gains in agriculture and food security, monthly transfers appeared more favorable for livestock accumulation, underscoring that each transfer schedule may serve different objectives and enjoy its own set of comparative advantages. The evidence generated indicates that lump-sum payments can deliver front-loaded gains in food security and agricultural investment among smaller scale farmers faced with time-sensitive and lumpy investment decisions, whereas monthly transfers may support gradual asset accumulation that can translate into livelihood diversification.

References

- Aggarwal, S., Aker, J. C., Jeong, D., Kumar, N., Park, D. S., Robinson, J., and Spearot, A. (2024). The dynamic effects of cash transfers to agricultural households. Technical report, National Bureau of Economic Research.
- Aiken, E., Bellue, S., Karlan, D., Udry, C., and Blumenstock, J. E. (2022). Machine learning and phone data can improve targeting of humanitarian aid. *Nature*, 603(7903):864–870.
- Aker, J. C., Boumnijel, R., McClelland, A., and Tierney, N. (2016). Payment mechanisms and antipoverty programs: Evidence from a mobile money cash transfer experiment in Niger. *Economic Development and Cultural Change*, 65(1):1–37.
- Alatas, V., Banerjee, A., Hanna, R., Olken, B. A., and Tobias, J. (2012). Targeting the poor: Evidence from a field experiment in Indonesia. *American Economic Review*, 102(4):1206–40.
- Baird, S., McIntosh, C., and Ozler, B. (2010). Cash or condition? Evidence from a cash transfer experiment. *World Bank Policy Research Working Paper*, (5259).
- Balantrapu, T., Christian, P., Dinarte-Diaz, L., Dunsch, F., Heirman, J., Jeong, D., Kelley, E., Kondylis, F., Lane, G., and Loeser, J. (2025). Does gender tagging public works increase women’s participation? Experimental evidence from Haiti, Kenya, and Rwanda. In *AEA Papers and Proceedings*, volume 115, pages 318–323.
- Balboni, C., Bandiera, O., Burgess, R., Ghatak, M., and Heil, A. (2022). Why do people stay poor? *Quarterly Journal of Economics*, 137(2):785–844.
- Baliki, G., Cruz, M., Dunsch, F., Garcia, J., Guariso, A., Heirman, J., and Weiffen, D. (2026). Targeting humanitarian aid in crisis: Evidence from the reduction of WFP assistance in Lebanon. Report, World Food Programme, Rome.
- Banerjee, A., Duflo, E., Goldberg, N., Karlan, D., Osei, R., Parienté, W., Shapiro, J., Thuysbaert, B., and Udry, C. (2015). A multifaceted program causes lasting progress for the very poor: Evidence from six countries. *Science*, 348(6236):1260799.
- Banerjee, A., Faye, M., Krueger, A., Niehaus, P., and Suri, T. (2020). Effects of a universal basic income during the pandemic. Working paper, Innovations for Poverty Action.
- Bastagli, F., Hagen-Zanker, J., Harman, L., Barca, V., Sturge, G., Schmidt, T., and Pellerano, L. (2016). Cash transfers: What does the evidence say? A rigorous review of programme impact and of the role of design and implementation features. Technical report, Overseas Development Institute (ODI), London, UK.

- Bationo, A., Fening, J., and Kwaw, A. (2018). *Improving the profitability, sustainability, and efficiency of nutrients through site-specific fertilizer recommendations in West Africa agro-ecosystems*, volume 1. Springer International Publishing. Chapter 7, Assessment of soil fertility status and integrated soil fertility management in Ghana.
- Bedoya, G., Coville, A., Haushofer, J., Isaqzadeh, M., and Shapiro, J. P. (2019). No household left behind: Afghanistan targeting the ultra poor impact evaluation. Technical report, National Bureau of Economic Research.
- Benhassine, N., Devoto, F., Duflo, E., Dupas, P., and Pouliquen, V. (2015). Turning a shove into a nudge? A “labeled cash transfer” for education. *American Economic Journal: Economic Policy*, 7(3):86–125.
- Berhane, G., Gilligan, D. O., Hoddinott, J., Kumar, N., and Taffesse, A. S. (2014). Can social protection work in Africa? The impact of Ethiopia’s productive safety net programme. *Economic Development and Cultural Change*, 63(1):1–26.
- Blattman, C., Fiala, N., and Martinez, S. (2014). Generating skilled self-employment in developing countries: Experimental evidence from Uganda. *Quarterly Journal of Economics*, 129(2):697–752.
- Christian, P., Dinarte, L., Dunsch, F. A., Heirman, J. L., Jeong, D., Kelley, E., Kondylis, F., Lane, G., Loeser, J. A., Uckat, H., Balantrapu, T., Fiorina, M.-A., McCollum, K., and Spindler, E. (2023). Impact evaluation of cash-based transfers on food security and gender equality in El Salvador. *Impact Evaluation of Cash-based Transfers on Food Security and Gender Equality in El Salvador. World Food Programme Office of Evaluation*.
- Crosta, T., Karlan, D., Ong, F., Rüschepöhler, J., and Udry, C. R. (2024). Unconditional cash transfers: A Bayesian meta-analysis of randomized evaluations in low and middle income countries. Technical report, National Bureau of Economic Research.
- Dercon, S. and Christiaensen, L. (2011). Consumption risk, technology adoption and poverty traps: Evidence from Ethiopia. *Journal of Development Economics*, 96(2):159–173.
- Fafchamps, M., McKenzie, D., Quinn, S., and Woodruff, C. (2014). Microenterprise growth and the flypaper effect: Evidence from a randomized experiment in Ghana. *Journal of Development Economics*, 106:211–226.
- Ghana Statistical Service (2022). Production of agriculture statistics, crop_region tab. Technical report, <https://statsghana.gov.gh/gssmain/fileUpload/Agriculture%20&%20Environment/AgricStatisticsPUB-2022Edition.xlsx> and https://statsghana.gov.gh/nationalaccount_macros.php?Stats=MjM3NTIyNzgZMy44ODU=/webstats/985rp49861.

- Ghana Statistical Service (2024). Statistical bulletin, Consumer Price Index (February, 2024). Technical report.
- Haushofer, J., Niehaus, P., Paramo, C., Miguel, E., and Walker, M. (2025). Targeting impact versus deprivation. *American Economic Review*, 115(6):1936–1974.
- Haushofer, J., Ringdal, C., Shapiro, J. P., and Wang, X. Y. (2019). Income changes and intimate partner violence: Evidence from unconditional cash transfers in Kenya. Technical report, National Bureau of Economic Research.
- Haushofer, J. and Shapiro, J. (2016). The short-term impact of unconditional cash transfers to the poor: Experimental evidence from Kenya. *The Quarterly Journal of Economics*, 131(4):1973–2042.
- Hidrobo, M., Hoddinott, J., Peterman, A., Margolies, A., and Moreira, V. (2014). Cash, food, or vouchers? Evidence from a randomized experiment in northern Ecuador. *Journal of Development Economics*, 107:144–156.
- Hoddinott, J., Sandström, S., and Upton, J. (2018). The impact of cash and food transfers: evidence from a randomized intervention in Niger. *American Journal of Agricultural Economics*, 100(4):1032–1049.
- Jeong, D. and Trako, I. (2022). Cash and in-kind transfers in humanitarian settings. *World Bank Development Impact Evaluation Group, Policy Research Working Paper*, 10026.
- Karlan, D., Osei, R., Osei-Akoto, I., and Udry, C. (2014). Agricultural decisions after relaxing credit and risk constraints. *Quarterly Journal of Economics*, 129(2):597–652.
- Karlan, D. and Thuysbaert, B. (2019). Targeting ultra-poor households in Honduras and Peru. *The World Bank Economic Review*, 33(1):63–94.
- Kondylis, F. and Loeser, J. (2024). Cash transfer size and persistence: Do larger cash transfers increase consumption more cost effectively? Working paper, Development Impact Evaluation Group, World Bank.
- McBride, L. and Nichols, A. (2018). Retooling poverty targeting using out-of-sample validation and machine learning. *The World Bank Economic Review*, 32(3):531–550.
- McIntosh, C. and Zeitlin, A. (2022). Using household grants to benchmark the cost effectiveness of a USAID workforce readiness program. *Journal of Development Economics*, 157:102875.
- Ministry of Food and Agriculture (2019). Agriculture in Ghana: Facts and figures (2018). Technical report.

- Ministry of Food and Agriculture and World Food Programme (2023). Food security and nutrition monitoring system - Ghana. Monthly Market Price Bulletin (July 2023). Technical report, <https://reliefweb.int/node/3998612>.
- Premand, P. and Schnitzer, P. (2021). Efficiency, legitimacy, and impacts of targeting methods: Evidence from an experiment in Niger. *The World Bank Economic Review*, 35(4):892–920.
- Schwab, B. (2019). Comparing the productive effects of cash and food transfers in a crisis setting: Evidence from a randomised experiment in Yemen. *The Journal of Development Studies*, 55(sup1):29–54.
- Tirivayi, N., Waidler, J., and Otchere, F. (2021). Cash transfers: Past, present and future. evidence and lessons learned from the transfer project. Technical report, UNICEF Office of Research – Innocenti, Florence, Italy.

A Appendix

A.1 Balance checks

Table A1: Balance tests and attrition

Variable	Lump Mean	Monthly Mean	Difference	P-value	N
Demographic and socio-economic characteristics					
Household size at registration	8.735	8.533	.202	.295	3237
Age of registered beneficiary	39.641	38.776	.865	.185	3237
Has some education (primary or above)	.371	.381	-.01	.676	3237
Female beneficiary	.543	.553	-.011	.677	3237
Agriculture-related characteristics					
Land size at registration (acres)	4.668	4.39	.278	.114	3237
Cultivates maize	.864	.869	-.005	.754	3237
Cultivates rice	.325	.351	-.026	.243	3237
Cultivates soy	.655	.673	-.018	.555	3237
Has own storage facility	.372	.35	.022	.293	3237
Engaged in the processing of ag output	.246	.223	.023	.326	3237
Is part of a cooperative	.044	.034	.01	.295	3237
Amount spent on ag inputs last season (GHS)	1977.553	1841.057	136.496	.145	3237
Amount planned for ag inputs this season (GHS)	2666.987	2515.419	151.568	.225	3237
Financial characteristics					
Is part of a savings group	.182	.193	-.011	.621	3237
Has bank account	.075	.09	-.015	.188	3237
Owned a business at registration	.097	.075	.022	.102	3237
Non-reponse relative to round 1					
Non-response in round 2	.077	.086	-.009	.526	3237
Non-response in round 3	.054	.07	-.016	.19	3237

Notes: The difference is the coefficient estimated based on regressing the listed variables on treatment (binary variable taking the value of 1 for the villages assigned to lump support) while controlling for the district of data collection and clustering errors at the village level. Round 1 sample of households. The Yes/No variables are binary coded. Besides maize, rice, and soya, registration data also included the rate of tomato, onion, and pepper cultivation, but they are not reported in the table. Less than 6–7% of farmers reported these crops during registration. Even fewer have ultimately reported cultivating these crops during the 2023 rainy season, i.e., below 1%. Continuous variables are winsorized at the 1st and 99th percentiles to correct outliers.

A.2 Outcome variables

Food (in)security

1. Food Consumption Score (FCS): is a weighted indicator based on the frequency of consumption of different food groups over the last seven days. There are 8 questions for 8 food groups, with 0–7 number of days as answer options. These are the food groups (along with their respective weights in parenthesis): cereals, grains, tubers, roots (2), legumes, pulses, nuts (3), milk and other dairy (4), meat, fish, eggs (4),

Table A2: Outcomes of interest

Outcome variables	Type	Round 1	Round 2	Round 3
Food Consumption Score (FCS)	Food security	✓	✓	✓
reduced Coping Strategies Index (rCSI)	Food security	✓	✓	✓
Food Insecurity Experience Scale (FIES)	Food security	✓	✓	✓
Patient Health Questionnaire (PHQ4)	Mental wellbeing	✓	✓	✓
Agricultural inputs and outputs	Crop cultivation			✓
Livestock-related activities	Other livelihoods		✓	✓
Business-related activities	Other livelihoods		✓	✓
Wage employment	Other livelihoods		✓	✓
Food and non-durables expenditure	Household finances		✓	✓
Durables expenditure	Household finances		✓	✓
Credit, savings	Household finances		✓	✓
Sample size		3237	2993	3054

Notes: Round 1 of data collection included fewer modules to allow the team to finalize the surveys before the second transfer took place.

vegetables and leaves (1), fruits (1), oil, fat, butter (0.5), and sugar, sweets (0.5). The range is 0 to 112, and a high FCS increases the likelihood that a household's food intake is adequate.

2. reduced Coping Strategies Index (rCSI): is a weighted indicator used to compare the hardship faced by households as measured by the frequency and severity of the food-related coping strategies they used when faced with food shortages. 5 questions with 0–7 number of days and various weights. The range is 0 to 56, with 56 being maximum food insecurity. These are the questions (along with their respective weights in parenthesis):

- In the past 7 days, as a result of not having enough food or money to buy food, did you have to rely on less preferred and less expensive food? If so, for how many days? (1)
- Borrow food or relied on help from friends or relatives? (2)
- Reduce portion size of meals? (1)
- Reduce the quantities consumed by adults/mothers so that young children can eat? (3)
- Reduce the number of meals eaten per day? (1)

3. Food Insecurity Experience Scale (FIES): is an experience-based measure of household food insecurity and consists of eight questions regarding the respondents' access to adequate food. There are 8 “yes” and “no” questions. The indicator is the sum

of answers. Thus, the range is 0 to 8, with 8 being maximum food insecurity. These are the questions:

- During the last 30 days, was there a time when you or others in your household worried about not having enough food to eat because of a lack of money or other resources?
- ... was there a time when you or others in your household were unable to eat healthy and nutritious food because of a lack of money or other resources?
- ... was there a time when you or others in your household ate only a few kinds of foods because of a lack of money or other resources?
- ... was there a time when you or others in your household had to skip a meal because there was not enough money or other resources to get food?
- ... was there a time when you or others in your household ate less than you thought you should because of a lack of money or other resources?
- ... was there a time when your household ran out of food because of a lack of money or other resources?
- ... was there a time when you or others in your household were hungry but did not eat because there was not enough money or other resources for food?
- ... was there a time when you or others in your household went without eating for a whole day because of a lack of money or other resources?

Mental health

Patient Health Questionnaire (PHQ4): is an ultra-brief screening tool for anxiety and depression. 4 questions with 0–3 gradation (0-Not at all, 1-Several days, 2-More than half the days, and 3-Nearly every day). The indicator is the sum of the answers, and the range is thus 0 to 12. PHQ4 scores can be categorized as normal (0–2), mild (3–5), moderate (6–8), and severe (9–12). The questions are:

- In the last 2 weeks, how often have you felt nervous, anxious or on edge?
- ... how often have you felt unable to stop or control worrying?
- ... how often have you felt down, depressed or hopeless?
- ... how often have you felt little interest or pleasure in doing things?

A.3 Robustness checks and additional results

Table A3: Variance of food security and farm scale heterogeneity

	(1)	(2)	(3)	(4)	(5)	(6)
	var(FCS)	var(FCS)	var(rCSI)	var(rCSI)	var(FIES)	var(FIES)
	(7 days)	(7 days)	(7 days)	(7 days)	(30 days)	(30 days)
Lump-sum payment	-16.813 (12.562)	13.819 (19.573)	-6.490 (6.269)	-10.194 (9.020)	-0.007 (0.250)	0.015 (0.349)
Farm scale registered (acres)		-4.864 (5.518)		-0.083 (2.171)		-0.066 (0.075)
Lump-sum $\times$ Farm scale		-6.393** (3.131)		1.101 (1.383)		-0.008 (0.063)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes
District $\times$ Farm scale	No	Yes	No	Yes	No	Yes
Observations	3136	3136	3136	3136	3136	3136
Mean variation	304	304	131	131	6	6

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). For each household with two or more observations, the variance of the three food security outcomes is computed. The recall period for each measure is between parentheses.

Table A4: Robustness to outlier correction for agricultural inputs

	(1) Area worked (Acres)	(2) Area worked (Acres, wins.)	(3) Area worked (Acres, ln)	(4) Pooled labour (Days)	(5) Pooled labour (Days, wins.)	(6) Pooled labour (Days, ln)
Panel A						
Lump-sum payment	0.197 (0.294)	0.327 (0.244)	0.057* (0.030)	4.693* (2.423)	4.391** (2.217)	0.189 (0.116)
Mean monthly group	6.08	5.90	1.77	22.41	22.07	1.93
Observations	3054	3054	3054	3054	3054	3054
	Chemical fertilizer (GHS)	Chemical fertilizer (GHS, wins.)	Chemical fertilizer (GHS, ln)	Organic fertilizer (GHS)	Organic fertilizer (GHS, wins.)	Organic fertilizer (GHS, ln)
Panel B						
Lump-sum payment	114.678 (75.790)	120.331** (59.759)	0.186** (0.090)	-8.021 (13.822)	0.664 (8.551)	0.020 (0.079)
Mean monthly group	1673.96	1628.64	6.52	59.25	42.47	0.49
Observations	3054	3054	3054	3054	3054	3054
	Seeds (GHS)	Seeds (GHS, wins.)	Seeds (GHS, ln)	Pesticides (GHS)	Pesticides (GHS, wins.)	Pesticides (GHS, ln)
Panel C						
Lump-sum payment	18.373 (27.867)	14.044 (23.936)	0.037 (0.132)	-2.880 (24.661)	-1.673 (20.562)	0.079 (0.079)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	372.76	361.16	4.13	410.44	396.57	5.36
Observations	3054	3054	3054	3054	3054	3054
	Hired labour (GHS)	Hired labour (GHS, wins.)	Hired labour (GHS, ln)			
Panel D						
Lump-sum payment	-23.091 (25.715)	-10.407 (18.379)	-0.074 (0.130)			
District dummies	Yes	Yes	Yes			
Mean monthly group	273.70	250.73	3.70			
Observations	3054	3054	3054			

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). For each outcome variable, the table shows the regressions for the raw values, winsorized values at the 1st and 99th percentiles, and the natural log. Data on inputs for the 2023 rainy season were collected once during round 3. Before taking the natural log, 1 is added to all outcome variables.

Table A5a: Robustness to outlier correction for agricultural harvest

	(1) Harvest (GHS)	(2) Harvest (GHS, wins.)	(3) Harvest (GHS, ln)
Panel A			
Lump-sum payment	-3252.009 (3758.645)	399.787 (393.385)	0.097 (0.067)
Mean monthly group	10682.72	6877.36	8.37
Observations	3054	3054	3054
	Harvest (Kg)	Harvest (Kg, wins.)	Harvest (Kg, ln)
Panel B			
Lump-sum payment	-909.338 (1036.348)	117.941 (107.519)	0.096 (0.065)
Mean monthly group	2876.63	1837.15	7.07
Observations	3054	3054	3054

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). For each outcome variable, the table shows the regressions for the raw values, winsorized values at the 1st and 99th percentiles, and the natural log. Data on inputs for the 2023 rainy season were collected once during round 3. Before taking the natural log, 1 is added to all outcome variables. Data on the inputs for the 2023 rainy season were collected once during round 3.

Table A5b: Robustness to outlier correction for agricultural sales

	(1) Sales (GHS)	(2) Sales (GHS, wins. 1-99)	(3) Sales (GHS, wins. 2-98)	(4) Sales (GHS, wins. 5-95)	(5) Sales (GHS, ln)	(6) Sales (GHS, sq root)	(7) Sales (GHS, cube root)
Panel A							
Lump-sum payment	308.891* (171.628)	282.995* (155.959)	280.051* (146.792)	266.523** (122.812)	0.316 (0.202)	3.521* (1.828)	0.769* (0.418)
Mean monthly group	1796.63	1755.92	1714.68	1592.75	5.19	31.38	8.61
Observations	3054	3054	3054	3054	3054	3054	3054
	Sales (Kg)	Sales (Kg, wins. 1-99)	Sales (Kg, wins. 2-98)	Sales (Kg, wins. 5-95)	Sales (Kg, ln)	Sales (Kg, sq root)	Sales (Kg, cube root)
Panel B							
Lump-sum payment	-464.373 (539.384)	85.524** (42.460)	82.271** (37.912)	75.142** (31.982)	0.268 (0.172)	0.988 (1.492)	0.408 (0.309)
Mean monthly group	1005.42	469.11	451.84	422.67	4.23	16.95	5.60
Observations	3054	3054	3054	3054	3054	3054	3054

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). For each outcome variable, the table shows the regressions for the raw values, winsorized values at various percentiles, the natural log, the square, and cube root. Data on inputs for the 2023 rainy season were collected once during round 3. Before taking the natural log, 1 is added to all outcome variables. Data on the inputs for the 2023 rainy season were collected once during round 3.

Table A6: Robustness to outlier correction for TLU, livestock profit, and consumption

	(1) TLU (Current)	(2) TLU (Current) (wins.)	(3) TLU (Current) (ln)
Panel A			
Lump-sum payment	-0.068 (0.142)	-0.176** (0.086)	-0.046* (0.024)
Mean monthly group	1.43	1.31	0.56
Observations	6047	6047	6047
	Consumed livestock (6 months) (GHS)	Consumed livestock (6 months) (GHS, wins.)	Consumed livestock (6 months) (GHS, ln)
Panel B			
Lump-sum payment	2.549 (17.007)	4.492 (10.575)	0.049 (0.089)
Mean monthly group	159.61	130.72	1.52
Observations	6047	6047	6047
	Profits livestock (6 months) (GHS)	Profits livestock (6 months) (GHS, wins.)	Profits livestock (6 months) (GHS, ln)
Panel C			
Lump-sum payment	13.741 (26.578)	4.561 (11.906)	0.033 (0.078)
Round dummies	Yes	Yes	Yes
District dummies	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes
Mean monthly group	193.28	144.36	1.36
Observations	6047	6047	6047

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). The winsorisation corrects outliers below the 1st and above the 99th percentiles. The winsorized TLU uses the winsorized stock of each type of animal and applies the appropriate weights before summing up to compute TLU. The log-normal TLU is computed by taking the natural log of the uncorrected TLU.

Table A7: TLU disaggregation

	(1) Poultry (stock)	(2) Goats (stock)	(3) Cows (stock)	(4) Sheep (stock)	(5) Pigs (stock)
Panel A: Round 2					
Lump-sum payment	0.024 (0.567)	0.066 (0.180)	-0.148* (0.085)	-0.170 (0.138)	-0.047 (0.031)
District dummies	Yes	Yes	Yes	Yes	Yes
Mean monthly group	7.86	2.97	0.77	1.65	0.17
Observations	2993	2993	2993	2993	2993
	Poultry (stock)	Goats (stock)	Cows (stock)	Sheep (stock)	Pigs (stock)
Panel B: Round 3					
Lump-sum payment	-0.486 (0.491)	0.071 (0.162)	-0.139 (0.087)	-0.314** (0.130)	-0.066* (0.037)
District dummies	Yes	Yes	Yes	Yes	Yes
Mean monthly group	8.99	2.98	0.78	1.75	0.23
Observations	3054	3054	3054	3054	3054
	Poultry (stock)	Goats (stock)	Cows (stock)	Sheep (stock)	Pigs (stock)
Panel C: Pooled					
Lump-sum payment	-0.234 (0.463)	0.068 (0.152)	-0.143* (0.075)	-0.242** (0.122)	-0.057* (0.030)
Round dummies	Yes	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes	Yes
Mean monthly group	8.43	2.98	0.78	1.70	0.20
Observations	6047	6047	6047	6047	6047

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). The winsorisation corrects outliers below the 1st and above the 99th percentiles.

Table A8a: Farm asset ownership per household

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Axes	Hoes	Machetes	Carts	Granaries	Yokes	Blades
Panel A: Round 2							
Lump-sum payment	0.041 (0.045)	0.089 (0.118)	0.005 (0.061)	0.014 (0.023)	-0.010 (0.021)	-0.010 (0.016)	-0.001 (0.093)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.93	3.97	1.83	0.16	0.14	0.18	1.19
Observations	2993	2993	2993	2993	2993	2993	2993
Panel B: Round 3							
Lump-sum payment	-0.040 (0.051)	0.075 (0.109)	-0.005 (0.059)	-0.011 (0.021)	-0.036 (0.025)	-0.032 (0.020)	0.003 (0.079)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	1.11	4.05	1.99	0.18	0.17	0.19	1.18
Observations	3054	3054	3054	3054	3054	3054	3054
Panel C: Pooled							
Lump-sum payment	0.000 (0.042)	0.082 (0.107)	-0.000 (0.056)	0.001 (0.021)	-0.023 (0.022)	-0.021 (0.016)	0.001 (0.080)
Round dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	1.02	4.01	1.91	0.17	0.16	0.18	1.18
Observations	6047	6047	6047	6047	6047	6047	6047

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the village level (163 clusters). The outcomes are measured at the household level, and all continuous variables are winsorized at the 1st and 99th percentiles to correct outliers.

Table A8b: Household asset ownership per household

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Tables	Chairs	Beds	Mattresses	Radios	TVs	Motorcycles	Bicycles	Phones	Msq. nets
Panel A: Round 2										
Lump-sum payment	0.004 (0.020)	-0.066 (0.106)	-0.004 (0.050)	-0.069 (0.069)	0.004 (0.029)	-0.046 (0.057)	-0.058* (0.035)	-0.088 (0.055)	-0.131 (0.089)	-0.011 (0.097)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.17	2.09	0.43	1.15	0.49	0.65	0.67	0.93	2.69	2.39
Observations	2993	2993	2993	2993	2993	2993	2993	2993	2993	2993
Panel B: Round 3										
Lump-sum payment	-0.033 (0.024)	-0.051 (0.102)	-0.016 (0.048)	0.009 (0.061)	-0.016 (0.028)	-0.016 (0.054)	-0.032 (0.034)	-0.095* (0.049)	-0.017 (0.074)	0.003 (0.085)
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.25	2.77	0.59	1.34	0.55	0.71	0.78	0.96	2.84	2.61
Observations	3054	3054	3054	3054	3054	3054	3054	3054	3054	3054
Panel C: Pooled										
Lump-sum payment	-0.015 (0.019)	-0.058 (0.098)	-0.010 (0.046)	-0.029 (0.062)	-0.006 (0.025)	-0.031 (0.054)	-0.045 (0.032)	-0.091* (0.050)	-0.073 (0.077)	-0.004 (0.085)
Round dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	0.21	2.44	0.51	1.24	0.52	0.68	0.73	0.94	2.77	2.50
Observations	6047	6047	6047	6047	6047	6047	6047	6047	6047	6047

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the village level (163 clusters). The outcomes are measured at the household level, and all continuous variables are winsorized at the 1st and 99th percentiles to correct outliers.

Table A9a: Farm scale heterogeneity and other livelihood activities

	Round 2				Round 3				Pooled			
	(1) TLU (current)	(2) Profits livestock (6 months)	(3) Profits business (30 days)	(4) Yearly earnings (12 months)	(5) TLU (current)	(6) Profits livestock (6 months)	(7) Profits business (30 days)	(8) Yearly earnings (12 months)	(9) TLU (current)	(10) Profits livestock (6 months)	(11) Profits business (30 days)	(12) Yearly earnings (12 months)
Lump-sum payment	-0.126 (0.169)	-36.546 (29.541)	2.834 (15.345)	-245.926 (401.580)	0.027 (0.129)	0.837 (24.214)	-3.239 (18.637)	525.303* (312.536)	-0.047 (0.133)	-10.736 (13.488)	-0.157 (14.215)	151.045 (295.313)
Farm scale registered (acres)	-0.007 (0.022)	5.090 (10.878)	-1.858 (6.581)	-61.164 (102.819)	0.008 (0.020)	11.914 (10.452)	7.642 (5.832)	-20.446 (76.009)	0.010 (0.018)	-1.341 (5.704)	1.560 (4.952)	-43.088 (67.704)
Lump-sum × Farm scale	-0.015 (0.034)	10.819* (5.594)	-1.903 (2.812)	34.417 (67.072)	-0.051** (0.024)	-2.430 (4.832)	-1.692 (3.032)	-78.055 (52.301)	-0.034 (0.025)	2.477 (2.552)	-1.837 (2.383)	-24.702 (48.154)
Round dummies	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
District dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District x Round dummies	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
District x Farm scale	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Round x Farm scale	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Observations	2993	2993	2906	2993	3054	3054	2990	3054	6047	9284	5896	6047
Marginal means at various percentiles farm scale												
Lump-Monthly at 5th perc.	-0.149	-20.317	-0.02	-194.3	-0.05	-2.809	-5.778	408.22	-0.098	-7.021	-2.913	113.992
Lump-Monthly at 25th perc.	-0.157	-14.907	-0.971	-177.092	-0.075	-4.024	-6.624	369.193	-0.115	-5.782	-3.831	101.641
Lump-Monthly at 50th perc.	-0.188**	6.732	-4.776	-108.259	-0.178*	-8.885	-10.009	213.083	-0.184**	-8.29	-7.506	52.236
Lump-Monthly at 75th perc.	-0.203**	17.551	-6.679	-73.842	-0.23**	-11.315	-11.702	135.028	-0.218***	1.648	-9.343	27.534
Lump-Monthly at 95th perc.	-0.28	71.648**	-16.191	98.242	-0.487***	-23.467	-20.164	-255.248	-0.388**	14.032	-18.53	-95.977

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). Variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. In terms of farm size distribution: 1.5 acres (5th percentile), 2 acres (25th percentile), 4 acres (50th percentile), 5 acres (75th percentile), and 10 acres (95th percentile).

Table A9b: Farm scale heterogeneity, assets, and finances

	Round 2				Round 3				Pooled			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Farm assets (Number)	Household assets (Number)	Balance savings (GHS)	Balance credit (GHS)	Farm assets (Number)	Household assets (Number)	Balance savings (GHS)	Balance credit (GHS)	Farm assets (Number)	Household assets (Number)	Balance savings (GHS)	Balance credit (GHS)
Panel A: Pooled												
Lump-sum payment	0.248 (0.375)	-0.271 (0.628)	-43.303* (22.391)	26.847 (50.977)	0.064 (0.341)	-0.305 (0.542)	1.013 (27.923)	-11.967 (60.267)	0.156 (0.336)	-0.292 (0.558)	-20.259 (20.347)	7.209 (44.242)
Farm scale registered (acres)	0.058 (0.077)	0.142 (0.109)	5.334 (8.623)	12.385 (10.453)	0.044 (0.066)	0.168 (0.121)	13.724 (9.488)	-6.712 (7.754)	0.043 (0.060)	0.174* (0.089)	7.669 (8.407)	-3.362 (6.906)
Lump-sum × Farm scale	-0.041 (0.059)	-0.079 (0.099)	3.208 (4.394)	-12.730* (7.538)	-0.043 (0.057)	-0.021 (0.084)	-4.958 (5.219)	-5.607 (12.990)	-0.043 (0.054)	-0.049 (0.085)	-1.101 (3.775)	-9.125 (8.002)
Round dummies	No	No	No	No	No	No	No	No	No	No	No	No
District dummies	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
District x Round dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean monthly group	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Observations	8.40	11.67	150.07	283.75	8.87	13.42	221.73	280.89	8.64	12.55	186.22	282.31
	2993	2993	2993	2993	3054	3054	3054	3054	6047	6047	6047	6047
Marginal means at various percentiles farm scale												
Lump-Monthly at 5th perc.	.187	-.389	-38.49**	7.752	-.001	-.337	-6.424	-20.376	.093	-.366	-21.911	-6.479
Lump-Monthly at 25th perc.	.166	-.428	-36.886**	1.387	-.023	-.348	-8.903	-23.18	.071	-.39	-22.461	-11.041
Lump-Monthly at 50th perc.	.084	-.586	-30.47**	-24.074	-.109	-.39	-18.819	-34.393	-.014	-.488	-24.664*	-29.292
Lump-Monthly at 75th perc.	.043	-.664	-27.262*	-36.804	-.152	-.411	-23.777	-39.999	-.056	-.537	-25.765*	-38.417
Lump-Monthly at 95th perc.	-.162	-1.058	-11.222	-100.455*	-.368	-.517	-48.567	-68.032	-.269	-.782	-31.271	-84.043

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Errors are clustered at the village level (163 clusters). All variables are winsorized. The winsorisation corrects outliers below the 1st and above the 99th percentiles. In terms of farm size distribution: 1.5 acres (5th percentile), 2 acres (25th percentile), 4 acres (50th percentile), 5 acres (75th percentile), and 10 acres (95th percentile).

Table A10: Household expenditure outcomes

	(1) Food expenditure (GHS)	(2) Non-food expenditure (GHS)
Panel A: Round 2		
Lump-sum payment	-9.896 (11.596)	24.366 (25.045)
Round dummies	Yes	Yes
District dummies	Yes	Yes
District x Round dummies	Yes	Yes
Mean monthly group	223.98	591.65
Observations	2993	2993
Panel B: Round 3		
Lump-sum payment	-5.662 (8.649)	3.025 (29.730)
District dummies	Yes	Yes
Mean monthly group	177.33	741.58
Observations	3054	3054
Panel C: Pooled		
Lump-sum payment	-7.758 (8.611)	13.589 (25.033)
District dummies	Yes	Yes
Mean monthly group	200.45	667.29
Observations	6047	6047

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the village level (163 clusters). All outcomes are measured at the household level. The outcome variables are as follows: (1) total most recent food-related purchases of food types consumed in the last 7 days, and (2) total *estimated* purchases on non-durable, non-food items per month. Outcomes are winsorized at the 1st and 99th percentiles to correct outliers.